4 Multiple Integrals

4.6 Triple Integrals

Definition 4.6.1 The triple integral of f over the box $B = [a, b] \times [c, d] \times [e, f]$ is given by

$$\iiint_B f(x, y, z)\, dV = \lim_{i,j,k\to} \sum_{i=1}^{\ell} \sum_{j=1}^{m} \sum_{k=1}^{n} f(x^*_{ijk}, y^*_{ijk}, z^*_{ijk}) \Delta V$$

Theorem 4.6.2 *If f is continuous on the box B, then*

$$\iiint_B f(x, y, z)\, dV = \int_a^b \int_c^d \int_e^f f(x, y, z)\, dz dy dx.$$

Example 4.6.3 Evaluate the integral $\iiint_B xyz^2\, dV$ where $B = [0, 1] \times [-1, 2] \times [0, 3]$.

Example 4.6.4 Evaluate the integral $\iiint_E z\, dV$, where E is the solid in the first octant bounded by the surface $z = 12xy$ and the planes $y = x$, and $x = 1$.

Example 4.6.5 Evaluate the integral $\iiint_E \sqrt{x^2 + z^2}\, dV$ where E is the region bounded by the paraboloid $y = x^2 + z^2$ and the plane $y = 4$.

Example 4.6.6 Use a triple integral to find the volume of the tetrahedron T bounded by the planes $x + 2y + z = 2$, $x = 2y$, $x = 0$ and $z = 0$.